



PHYS 551 – Advanced Monte Carlo Methods

Spring 2025/26

TERM PROJECT

**Rare-Event Estimation of Cascading Blackouts in the
Istanbul Power Grid via Importance Sampling**

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1 Introduction

A large electrical grid does not fail one component at a time. When a single district is pushed past the limit of its substations, its excess load does not simply vanish; it is rerouted onto the neighbouring districts. If those neighbours are themselves close to their limit, the overload spreads, and a local fault grows into a city-wide blackout. This chain reaction is called a *cascading failure*. Because the number of possible failure combinations in a grid of tens of districts is enormous, and because the truly catastrophic events are very rare, deterministic worst-case analysis is not practical. This term project therefore studies the problem with Monte Carlo simulation.

The grid studied here is the mainland of Istanbul, modelled as 38 districts on the European and Asian sides. Each district has a population, a daily electricity load, and a safety margin that is calibrated from real overload records published by EPIAŞ. The central difficulty is that a blackout affecting millions of people is a *rare event*: under normal operation it almost never happens, so a naive (crude) Monte Carlo simulation would have to run for an impractically long time before it ever observed one. The project answers two questions:

1. What is the probability that a cascading failure leaves at least X people without power on a given day, including the rare tail that naive Monte Carlo cannot reach? I answer this with **Importance Sampling**.
2. Given the resulting risk map, *where* should new relief substations be built to reduce the expected number of people affected the most?

The remainder of this report is organised as follows. Section 2 describes the data and the model: the districts, how loads and safety margins are obtained, and the cascading-failure mechanism. Section 3 presents the naive Monte Carlo baseline and shows where it breaks down. Section 4 introduces the Importance Sampling estimator and the systemic-surge bias that makes the rare tail tractable. Section 5 compares the two methods directly through the exceedance curve, the variance-reduction table, and a convergence study. Section 6 applies the model to the substation-placement question. Section 7 concludes.

Declaration of AI Usage

In compliance with the course’s principles for the responsible use of AI, I disclose my use of AI tools in this project. The Python code was written by an AI assistant following my prompts and direction. The core of the work, however, is my own: the design of the algorithm and how the model works – the cascading-failure mechanism, the systemic-surge demand model, and the Importance Sampling scheme – together with the calibration choices and the sourcing of the real data (the EPIAŞ electricity-consumption and overload “Aşırı Yük” records that set the district loads and safety margins) were all done by me. I directed the AI at each step and reviewed and verified every result against the simulation before including it.

2 System Model and Data

2.1 Districts, Population, and Load

The grid is represented as a graph of 38 mainland districts (the islands are excluded), with a total population of about 15 736 564. Two districts are connected by an edge if they share a physical border, giving 83 edges; the adjacency was extracted from district geometry and adjusted for the Bosphorus bridge crossings. Figure 1 shows the resulting network.

Each district is assigned a base electricity load μ_i . The total monthly consumption of the European and Asian distribution regions (from EPIAŞ consumption data) is distributed among

the districts of each side in proportion to their population:

$$\mu_i = C_{\text{side}} \cdot \frac{\text{Pop}_i}{\sum_{j \in \text{side}} \text{Pop}_j}, \quad (1)$$

where C_{side} is the average load of the side that district i belongs to. This makes the load distribution reflect the real demand split between the two sides of the city.

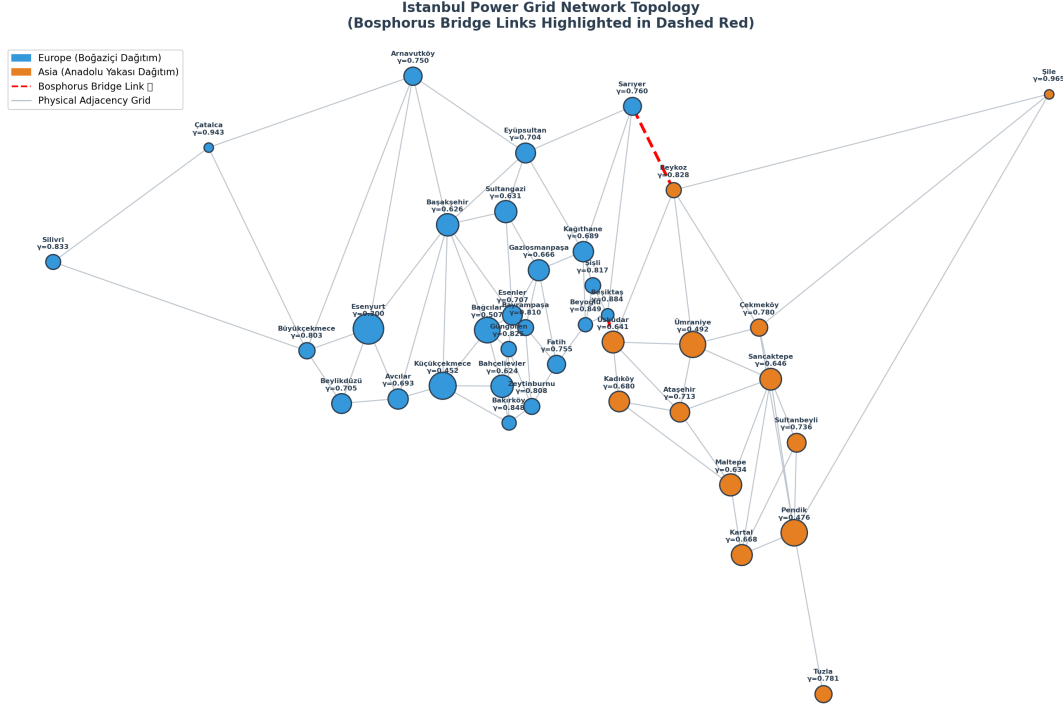


Figure 1: The Istanbul mainland grid as a graph of 38 districts. Nodes are placed at district centroids; edges connect bordering districts.

2.2 Safety Margins from Real Overload Data

Each district is given a dimensionless safety margin α_i , defined so that its capacity is

$$\text{capacity}_i = \mu_i (1 + \alpha_i). \quad (2)$$

A district with a large α_i can absorb a big demand spike before tripping; a district with a small α_i is fragile. Rather than assign these margins arbitrarily, they are calibrated from one year of EPIAŞ outage records by counting, for each district, how many outages were caused by overload (“Aşırı Yük”). The margins are then assigned *per side* (because the two sides are operated by different distribution companies) by linearly mapping the overload count to the range $[\alpha_{\min}, \alpha_{\max}] = [0.08, 0.25]$:

$$\alpha_i = \alpha_{\max} - \frac{n_i}{\max_{j \in \text{side}} n_j} (\alpha_{\max} - \alpha_{\min}), \quad (3)$$

where n_i is the overload count of district i . The districts with the most recorded overloads are therefore given the smallest margins. For example, Maltepe (465 overloads, the worst on the Asian side) and Fatih and Beyoğlu (108 each, the worst on the European side) receive $\alpha = 0.08$, while a quiet district such as Çatalca (2 overloads) is close to $\alpha = 0.25$. This is what makes the model data-driven: the historically weakest districts are the structurally weakest in the simulation.

2.3 The Cascading-Failure Mechanism

Daily demand. Electricity demand fluctuates from day to day. The demand of district i on a given day is modelled as

$$L_i = \mu_i (S + \varepsilon_i), \quad (4)$$

where $S \sim \mathcal{N}(1, \sigma_{\text{sys}}^2)$ is a single **city-wide demand multiplier** shared by every district, and $\varepsilon_i \sim \mathcal{N}(0, \sigma_{\text{idio}}^2)$ is a small district-specific fluctuation. The systemic term S represents a common cause, such as a heatwave that turns on every air conditioner in the city at once; it is the reason failures are correlated, and a rare large value of S is what drives a city-wide cascade. I use $\sigma_{\text{sys}} = 0.03$ and $\sigma_{\text{idio}} = 0.02$.

Tripping and redistribution. A district trips when its demand exceeds its capacity, i.e. when $L_i > \text{capacity}_i$. When district i trips, a fraction of its excess load is pushed onto its still-active neighbours:

$$\text{load to move} = \gamma_i (L_i - \text{capacity}_i), \quad (5)$$

where $\gamma_i \in (0, 1)$ is a redistribution factor. The load is split among the live neighbours in proportion to each neighbour’s remaining headroom, so load preferentially flows to the districts most able to absorb it. A neighbour that is itself pushed over its capacity then trips in turn, and the process repeats until no further districts fail. The redistribution factor is kept deliberately moderate ($\gamma_i \leq 0.35$) so that a single trip does not automatically collapse the whole city; a truly large blackout requires a large systemic surge S , which is rare. The quantity of interest at the end of each simulated day is the total **population without power**, i.e. the sum of the populations of all tripped districts.

3 Naive Monte Carlo Baseline

The naive (crude) estimator samples the model exactly as described above: for each of N simulated days, a systemic surge S and the district fluctuations ε_i are drawn from their true distributions, the cascade is run, and the affected population is recorded. The probability that at least X people lose power is estimated by the fraction of days on which this happened,

$$\hat{P}(\text{affected} \geq X) = \frac{1}{N} \sum_{k=1}^N \mathbf{1}\{\text{affected}_k \geq X\}, \quad (6)$$

with the usual binomial standard error $\sqrt{\hat{P}(1 - \hat{P})/N}$.

Table 1 shows the result of a run with $N = 100\,000$ days. The estimator is accurate for the common, small events: outages affecting more than 100 000 people occur on about 5.6% of days, and the curve is well-resolved down to the 4 000 000-person level (45 observed events). But the rarest bucket reveals the central problem: in 100 000 simulated days, **not a single** event affecting more than 8 000 000 people was observed. The naive estimate there is exactly zero, with only a useless “rule-of-three” upper bound of $3/N = 3 \times 10^{-5}$. The method is blind to the very events that matter most.

Table 1: Naive Monte Carlo exceedance probabilities, $N = 100\,000$ days.

Threshold X (people)	Observed events	$\hat{P}(\text{affected} \geq X)$
100 000	5603	5.60×10^{-2}
1 000 000	1466	1.47×10^{-2}
2 000 000	214	2.14×10^{-3}
4 000 000	45	4.50×10^{-4}
6 000 000	6	6.00×10^{-5}
8 000 000	0 not observed ($< 3 \times 10^{-5}$)	

4 Importance Sampling

4.1 Idea and Estimator

Importance Sampling removes the blindness of the naive method by drawing samples from a different, biased distribution g that makes the rare event common, and then correcting for the bias with a weight. If the true distribution is f and I instead sample from a proposal g , the unbiased estimator of any probability is

$$\hat{P} = \frac{1}{N} \sum_{k=1}^N \mathbf{1}\{\text{event}_k\} W_k, \quad W_k = \frac{f(x_k)}{g(x_k)}. \quad (7)$$

The weight W_k compensates for the fact that the biased distribution visits the rare region more often than reality would.

4.2 Biasing the Systemic Surge

The key modelling insight is that, in this grid, almost all of the tail risk is driven by the single systemic variable S : a city-wide blackout happens essentially only when S is large. I therefore bias *only* S , shifting its mean upward by θ standard deviations,

$$g: S \sim \mathcal{N}(1 + \theta \sigma_{\text{sys}}, \sigma_{\text{sys}}^2), \quad (8)$$

while the district fluctuations ε_i and the random tripping are left untouched (they are common to both methods and need no reweighting). Because the bias acts on a single dimension, the likelihood ratio has a clean closed form,

$$\ln W = -\theta z_S + \frac{1}{2}\theta^2, \quad z_S = \frac{S - 1}{\sigma_{\text{sys}}}, \quad (9)$$

and the weights stay well-behaved. This is far more stable than biasing all 38 districts independently, which would make the weights degenerate.

A single value of θ samples one part of the tail well. To cover the whole exceedance curve, a small *ladder* of tilts $\theta \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$ is run with 80 000 samples each, and each exceedance threshold is estimated from whichever tilt resolves it most precisely. The rarest bucket ($\geq 8\,000\,000$ people), which the naive method never observed, is now estimated as

$$\hat{P}(\text{affected} \geq 8\,000\,000) = (7.5 \pm 2.4) \times 10^{-6}, \quad (10)$$

i.e. roughly a one-in-133,000-day event, with a finite and usable confidence interval.

5 Comparison of the Two Methods

5.1 The Exceedance Curve

Figure 2 overlays the two estimators. On the common events (left side of the curve) the naive and Importance Sampling estimates lie exactly on top of each other – this agreement is the essential check that the Importance Sampling estimator is unbiased, since a wrong implementation would disagree here. As the events become rarer, the naive error bars grow, and at the 8 000 000-person level the naive method has no estimate at all (marked “blind”), whereas Importance Sampling continues smoothly down to 10^{-6} .

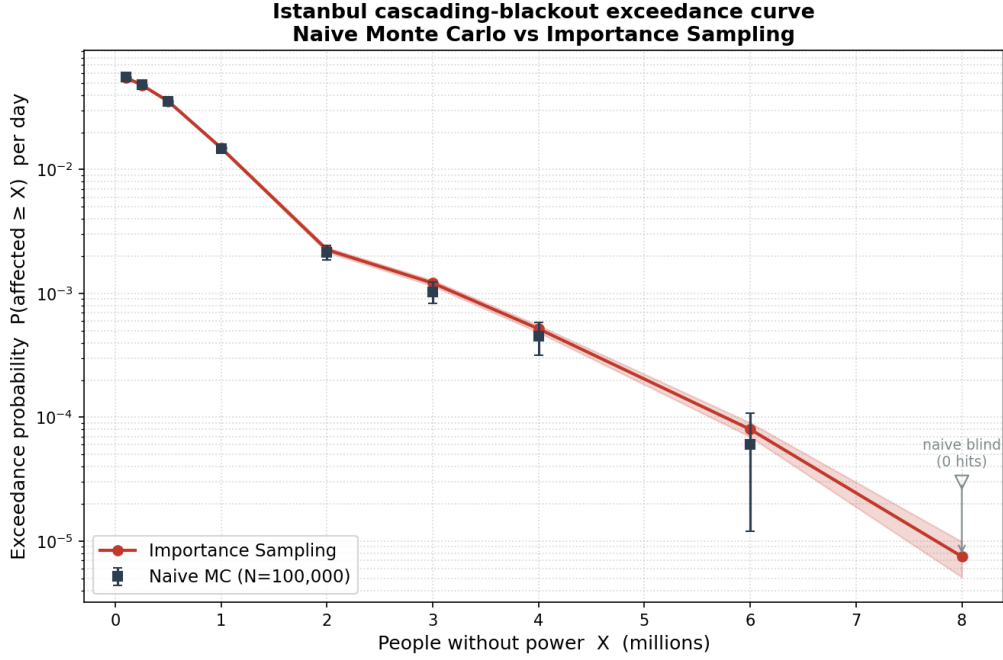


Figure 2: Exceedance probability $P(\text{affected} \geq X)$ per day. The two methods agree on common events; only Importance Sampling reaches the rare tail, where the naive estimator is blind (0 events observed).

5.2 Variance Reduction

Table 2 quantifies the gain. The variance-reduction factor is the ratio of the per-sample variances of the two estimators, evaluated at the true probability. For common events the gain is modest (about $2\times$), but it grows steadily as the event becomes rarer, reaching about $63\times$ for the rarest bucket. The final column states the gain in the most intuitive way: to observe even a *single* $\geq 8\,000\,000$ event, the naive method would need on the order of 133 000 simulated days on average, whereas Importance Sampling estimates its probability from only 80 000 biased samples.

Table 2: Per-threshold comparison of naive Monte Carlo and Importance Sampling, and the resulting variance-reduction factor.

Threshold X	Naive $\hat{P}$	IS $\hat{P}$	Var. reduction	Naive days / event
100 000	5.60×10^{-2}	5.53×10^{-2}	$2\times$	18
1 000 000	1.47×10^{-2}	1.49×10^{-2}	$6\times$	67
2 000 000	2.14×10^{-3}	2.25×10^{-3}	$10\times$	444
4 000 000	4.50×10^{-4}	5.18×10^{-4}	$22\times$	1931
6 000 000	6.00×10^{-5}	7.99×10^{-5}	$36\times$	12 512
8 000 000	blind (0)	7.51×10^{-6}	$63\times$	133 079

5.3 Convergence

Figure 3 tracks the running estimate of the rarest bucket against the number of samples. The Importance Sampling estimate settles onto its value of about 7.5×10^{-6} with a shrinking confidence band, while the naive estimate stays pinned at zero, its only information being an upper bound that falls slowly as $3/N$. This is a direct picture of why the rare-event problem requires variance reduction rather than simply more samples.

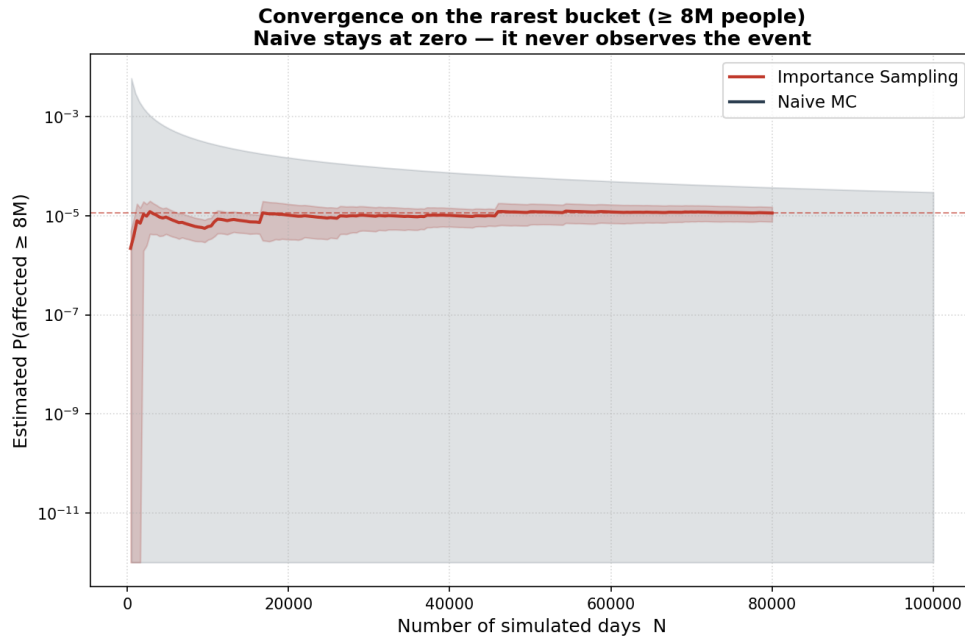


Figure 3: Convergence of the estimate for $P(\text{affected} \geq 8\,000\,000)$. Importance Sampling converges; the naive estimate never leaves zero.

5.4 Where the Risk Lies

Combining the per-district failure probabilities with the populations gives the expected number of people without power on an average day, which is about 44 000. Figure 4 maps the per-district blackout probability. The risk is strongly concentrated: a handful of low-margin districts dominate, led by Fatih (2.70%), Beyoğlu (2.60%), and Maltepe (2.57%) – exactly the districts that the EPIAŞ data flagged as having the most overloads. Most of the city is green (negligible risk). This concentration is what makes targeted reinforcement, studied next, so effective.

Table 3: Effect of the two selected relief substations.

Metric	Baseline	With 2 stations	Improvement
Expected daily affected (people)	43 991	1051	-97.6%
$P(\text{affected} \geq 1\,000\,000)$	1.50×10^{-2}	1.90×10^{-5}	$\sim 790\times$
$P(\text{affected} \geq 8\,000\,000)$	8.45×10^{-6}	1.53×10^{-7}	$\sim 55\times$

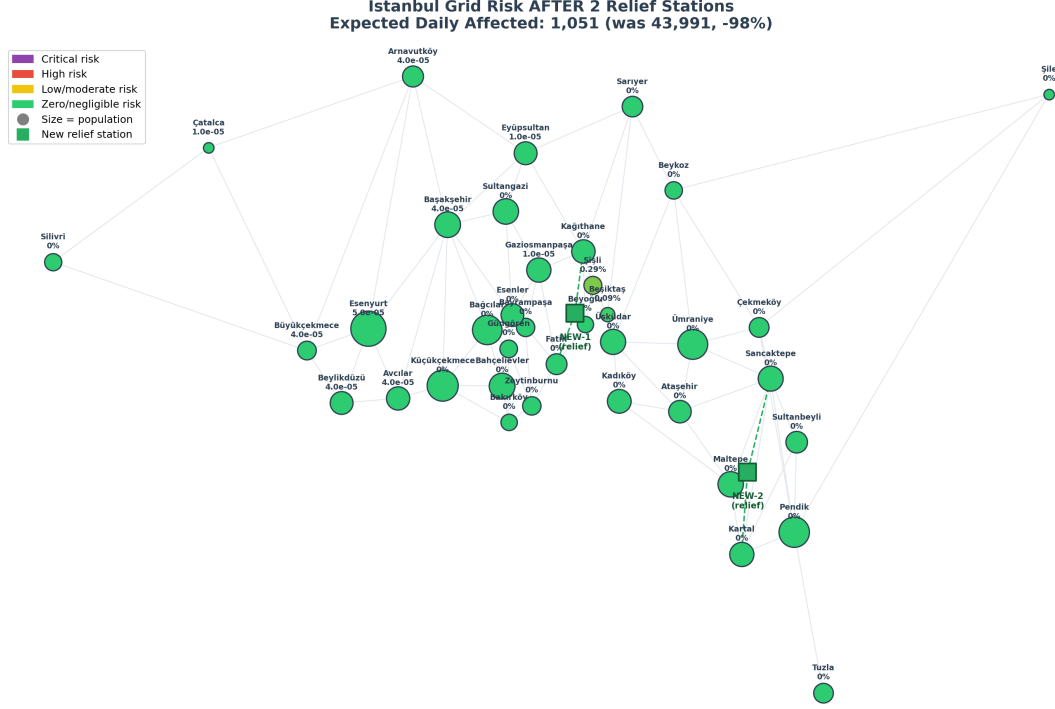


Figure 5: Risk map after adding the two relief substations (green squares). The previously critical clusters are neutralised and the city is almost entirely low-risk.

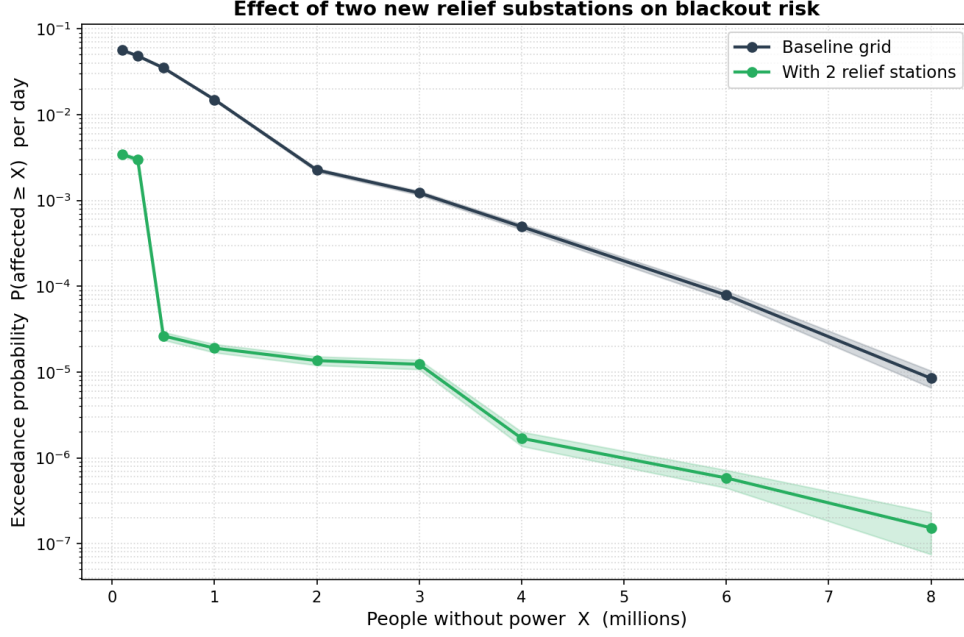


Figure 6: Exceedance curves before and after adding the two relief substations. The curve drops by orders of magnitude across all event sizes.

7 Conclusion

This project modelled cascading blackouts in the Istanbul mainland grid as a Monte Carlo problem, with district safety margins calibrated from real EPİAŞ overload data. The central finding is methodological: a city-wide blackout is a genuinely rare event, and naive Monte Carlo is blind to it – in 100 000 simulated days it never once observed an event affecting more than 8 million people. By biasing a single systemic demand variable and reweighting, Importance Sampling estimated the same probability as $(7.5 \pm 2.4) \times 10^{-6}$, with a variance reduction of up to about 63×, while agreeing exactly with the naive method on the common events that validate it. Finally, the risk map showed that vulnerability is concentrated in a few historically weak districts, and a greedy placement of two relief substations on those clusters cut the expected affected population by about 98%. The work demonstrates both why variance reduction is necessary for rare-event risk and how the resulting estimates can directly inform an engineering decision.